

# Do Game-Theoretic Decision Rules Improve Two-Voice Music Generation?

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**Abstract**—This paper studies whether game-theoretic decision rules improve two-voice symbolic music generation beyond a strong learned joint-compatibility baseline. We extract soprano and bass voices from JSB Chorales and train a multi-head utility model that predicts upper- and lower-voice private scores together with a shared upper-lower compatibility score. We compare independent generation, a no-game joint baseline, compatibility-filtered QRE-style generation, soft Stackelberg leader-follower generation, and a pure Nash equilibrium ablation. Preliminary analysis shows that strategically stable solvers can amplify sustained-note behavior, motivating a duration-aware state representation with explicit hold-length and onset-distance features. After selecting solver settings on validation data, we evaluate one matched 128-step continuation for each of all 77 held-out test chorales and quantify cross-chorale uncertainty with 10,000 paired bootstrap resamples. No-game joint retains the lowest harmonic interval-class and voice-distance JS point estimates, while QRE obtains the lowest hold-ratio and melodic-interval point estimates but none of these apparent advantages over No-game excludes zero under paired bootstrap. Both Stackelberg variants reliably improve local 3-gram diversity matching relative to No-game, while upper-leading Stackelberg shows no reliable difference from No-game on the two main vertical-distribution metrics. Pure Nash robustly collapses toward excessive sustain and very low diversity. A blind A/B study with 47 participants, using validation-selected QRE and Lower Stackelberg configurations, generally favors No-game joint, most clearly for perceived coherence over Lower Stackelberg. Overall, game-theoretic inference meaningfully changes generation behavior, but its benefits are selective, metric-dependent, and not automatically perceptual.

**Index Terms**—symbolic music generation, multi-agent generation, game theory, QRE, Stackelberg game, Nash equilibrium, JSB Chorales, two-voice music

## I. INTRODUCTION

Symbolic music generation is often formulated as a next-token prediction task. In multi-voice music, however, musical quality depends not only on the plausibility of each individual voice but also on the interaction between voices. In two-voice writing, the upper voice should form a coherent melodic line, the lower voice should provide a plausible bass trajectory, and the two voices should jointly maintain natural vertical intervals, consonance patterns, and voice distances.

This work asks whether two-voice generation can benefit from a game-theoretic decision process. At each time step, an upper-voice agent and a lower-voice agent select the next so-

prano and bass tokens. A learned utility model provides private scores for each voice and a shared compatibility score for each candidate pair. A decision rule then selects the next two-voice token pair from a common candidate grid. In musical terms, these decision rules represent different assumptions about how the two voices coordinate. The no-game joint baseline asks whether a learned vertical compatibility score is sufficient. The compatibility-filtered quantal response equilibrium (QRE) rule asks whether both voices should react probabilistically to each other’s utilities. The Stackelberg rule asks whether one voice should guide the other through a leader-follower structure. The Nash rule asks whether locally stable decisions are musically useful.

The central question is:

Does a game-theoretic decision rule improve two-voice generation beyond a no-game learned joint-compatibility baseline under the same model, state representation, candidate set, seeds, and continuation length?

A key contribution of this study is the inclusion of a strong no-game joint baseline. Without this baseline, an improvement over independent voice generation could be incorrectly attributed to game solving, even if the improvement mainly comes from modeling upper-lower compatibility. We therefore separate two effects: (i) whether modeling joint compatibility helps, and (ii) whether a game solver adds value beyond learned compatibility.

Our results give a nuanced answer. Learned joint compatibility remains a strong vertical-coordination mechanism, but the expanded full-test evaluation shows that the game solvers are not merely redundant. QRE shifts duration and melodic statistics, although its apparent advantages over No-game on those metrics are not reliable under paired bootstrap. Both Stackelberg leader variants produce a repeatable improvement in local 3-gram diversity matching, while upper-leading Stackelberg remains very close to No-game on harmonic interval-class and voice-distance distributions. Pure Nash equilibrium, despite local strategic stability, robustly collapses toward excessive sustain. The listening test adds a complementary perceptual result: the validation-selected Lower Stackelberg variant improves local-diversity matching objectively, yet lis-

teners still favor No-game joint for perceived coherence. The main lesson is therefore not that game theory is useless or uniformly beneficial, but that strategic decision rules induce specific musical trade-offs whose value depends on the metric and on human perception.

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces the dataset and representation. Section IV describes the utility model and generation rules. Sections V–VII present the experimental setup, validation results, and final objective test results. Section VIII reports the listening test. Finally, Sections IX–XI discuss the results, limitations, and conclusions.

## II. RELATED WORK

### A. Symbolic Polyphonic and Chorale Generation

Symbolic polyphonic music generation has often been studied as a sequence modeling problem. Boulanger-Lewandowski et al. [1] modeled temporal dependencies in high-dimensional polyphonic music sequences and helped establish JSB Chorales as a common benchmark for symbolic music generation. Our work also uses JSB Chorales, but focuses on a simplified two-voice setting by extracting soprano and bass voices in order to study explicit upper–lower interaction.

Several systems have specifically targeted Bach-style chorale or counterpoint generation. DeepBach [2] generates four-part Bach chorales using a steerable probabilistic neural model and pseudo-Gibbs sampling. Coconet, introduced in Counterpoint by Convolution [3], treats counterpoint generation as partial score completion and uses convolutional modeling with blocked Gibbs sampling. These systems demonstrate that polyphonic music generation benefits from modeling vertical and horizontal musical dependencies. However, their main focus is high-quality chorale or counterpoint generation through neural architectures and sampling procedures, rather than controlled comparison of game-theoretic decision rules.

Beyond chorale-specific systems, MuseGAN [4] studies symbolic multi-track generation with generative adversarial networks, emphasizing that multiple musical tracks have their own temporal dynamics while unfolding interdependently. This is related to our two-voice setting because both tasks require coordination across simultaneous musical parts. The difference is that MuseGAN learns a generative model for multi-track piano-roll generation, while our work keeps a learned utility model fixed and studies how different inference-time decision rules coordinate two voices.

### B. Reward-Guided and Multi-Agent Music Generation

Another related direction modifies or guides generation with rewards, constraints, or agent structures. RL Tuner [5] fine-tunes an RNN music generator with reinforcement learning rewards based on music-theoretic rules and learned model scores. This shares our motivation of going beyond ordinary next-token prediction, but differs in where the additional structure is applied: RL Tuner changes the trained policy through reinforcement learning, while our method changes the inference-time decision rule using learned utilities.

Recent multi-agent symbolic music systems also highlight the usefulness of decomposing musical creation into interacting roles. ComposerX [6] uses multiple LLM-based agents for symbolic music composition and reports that a multi-agent approach improves GPT-4’s composition quality. This is conceptually related to our use of upper and lower voices as separate agents. However, ComposerX operates at a high-level compositional workflow, whereas our agents correspond directly to musical voices and make local token-level decisions at each timestep.

Multi-agent reinforcement learning has also been explored for structured symbolic music generation [7]. Such work uses agent-based learning to address musical structure and long-range organization. In contrast, our system is not trained with reinforcement learning. We train a supervised multi-head utility model, then compare QRE-style, Stackelberg, and Nash decision rules at generation time.

### C. Game-Theoretic Interaction in Music

Game theory has a long history of modeling strategic interaction, including Nash equilibrium [8], quantal response equilibrium [9], and Stackelberg leader–follower games [10]. These concepts have been widely used in economics and multi-agent decision making, but their role in music generation remains relatively underexplored.

A recent example connecting music and game theory is reinforcement learning jazz improvisation [11], which models duo jazz improvisation as a game and studies improvisational strategies with reinforcement learning. This work is related in spirit because it treats musical interaction as a strategic process. Our setting is different: we study symbolic two-voice chorale-style generation, and we apply game-theoretic decision rules locally to learned soprano–bass utility matrices.

Therefore, the gap addressed by this paper is not simply “using multiple voices” or “using agents” for music generation. Prior work has already modeled polyphonic music, multi-track dependencies, reward-guided generation, and multi-agent composition. Our contribution is a controlled evaluation of classical game-theoretic decision rules as inference-time mechanisms for coordinating two symbolic musical voices. By using the same learned utility model, candidate grid, opening seeds, and continuation length across methods, we test whether QRE-style, Stackelberg, or Nash-based selection adds value beyond a strong no-game joint compatibility baseline.

## III. DATASET AND REPRESENTATION

### A. Dataset

We use the JSB Chorales dataset in a voice-separated sixteenth-note representation, following prior work on symbolic polyphonic music modeling [1]. Each timestep contains four aligned SATB voices:

$$[\text{soprano}, \text{alto}, \text{tenor}, \text{bass}]. \quad (1)$$

This project focuses on two-voice interaction, so we extract:

$$[\text{soprano}, \text{bass}], \quad (2)$$

where soprano is treated as the upper voice and bass as the lower voice.

The processed data are stored as variable-length token sequences. We use the `music21` toolkit for symbolic music handling and MusicXML export [12]. The split sizes used in our experiments are:

$$\begin{aligned} &229 \text{ train pieces, } 76 \text{ validation pieces, } 77 \text{ test pieces.} \\ &\quad (3) \end{aligned}$$

### B. Tokenization

The representation uses a sixteenth-note grid. Consecutive repeated non-rest pitches are converted into a hold token to distinguish sustained notes from repeated onsets:

- -2: HOLD,
- -1: REST,
- 0-127: MIDI pitch.

Repeated rests remain REST rather than HOLD. This distinction is important because a sustained pitch and a newly articulated repeated pitch have different musical meanings.

## IV. METHOD

### A. Original and Duration-Aware State

The original state representation contains a fixed two-voice context and bar position:

$$s_t = \{x_{t-k:t-1}, p_t\}, \quad (4)$$

where  $x_{t-k:t-1}$  is the previous  $k$  two-voice token context and  $p_t = t \bmod 16$  is the position in the bar. In all experiments,  $k = 16$ .

Preliminary analysis shows that stable game solvers, especially Stackelberg and Nash, can amplify HOLD behavior. Although HOLD tokens appear in the context, the model is not explicitly told how long the current note has been sustained or how long it has been since the last onset. We therefore introduce a duration-aware state representation:

$$s_t = \{x_{t-k:t-1}, p_t, d_t\}, \quad (5)$$

where  $d_t$  contains six duration features:

$$\text{upper current hold length,} \quad (6)$$

$$\text{lower current hold length,} \quad (7)$$

$$\text{upper steps since last onset,} \quad (8)$$

$$\text{lower steps since last onset,} \quad (9)$$

$$\text{upper previous token is HOLD,} \quad (10)$$

$$\text{lower previous token is HOLD.} \quad (11)$$

The first four features are normalized by 16. The feature dimension is therefore 6. To keep the comparison fair, all duration-aware generation methods use the same duration-aware checkpoint and the same state representation.

### B. Multi-Head Utility Model

We train a multi-head MLP with three output heads:

$$V_U(s_t, a_U) : \text{upper-voice private score,} \quad (12)$$

$$V_L(s_t, a_L) : \text{lower-voice private score,} \quad (13)$$

$$C(s_t, a_U, a_L) : \text{shared pair compatibility.} \quad (14)$$

In implementation, these are the soprano logits, bass logits, and pair logits. The model is trained with cross-entropy losses for upper-token prediction, lower-token prediction, and joint-pair prediction:

$$\mathcal{L} = \mathcal{L}_{upper} + \mathcal{L}_{lower} + \mathcal{L}_{pair}. \quad (15)$$

The model does not use a ranking loss or explicit negative sampling.

### C. Candidate Grid

Let  $V$  denote the token vocabulary built from the training split. At each generation step, we construct top- $k$  voice-level candidate sets:

$$A_U(s_t) = \text{top-}k \text{ tokens according to } V_U, \quad (16)$$

$$A_L(s_t) = \text{top-}k \text{ tokens according to } V_L. \quad (17)$$

The interacting methods choose from the shared grid:

$$A_U(s_t) \times A_L(s_t). \quad (18)$$

We use  $k = 12$ , giving 144 candidate pairs per step. This grid is shared by No-game joint, compatibility-filtered quantal response equilibrium (QRE), Stackelberg, and Nash methods.

### D. Generation Rules

Before introducing the individual methods, we group the decision rules by the type of coordination assumption they impose. The no-game joint baseline uses only the learned pair-compatibility score, so it tests whether ordinary learned vertical compatibility is already sufficient for coordination. Compatibility-filtered quantal response equilibrium (QRE) is a simultaneous soft-response rule: each voice probabilistically favors higher-utility actions rather than making a hard best response. Stackelberg generation is a leader-follower rule: one voice is treated as the leader and the other responds. Pure Nash generation is the strictest local stability rule: the selected pair must be one where neither voice can improve by changing its own token alone. This progression, from compatibility-only selection to soft response, leader-follower response, and strict equilibrium, lets us test whether stronger strategic structure improves music generation or instead makes the output overly conservative.

1) *Independent Baseline*: The independent baseline samples upper and lower tokens independently from their private heads. It does not use pair compatibility or a game solver.

2) *No-game joint Compatibility Baseline*: The no-game joint baseline uses the same candidate grid as the game solvers, but samples using only the shared compatibility score  $C(s_t, a_U, a_L)$ . It models upper-lower interaction but does not construct player-specific utilities or solve a game. This is the strongest non-game baseline.

3) *QRE*: QRE-style generation represents a soft simultaneous interaction. Instead of assuming that each voice always chooses a hard best response, each voice assigns higher probability to actions with higher utility. This is useful for music generation because deterministic best responses may be too rigid, while stochastic soft responses preserve variation.

The QRE-style method constructs two player-specific utility matrices:

$$U_U(s_t, a_U, a_L) = C(s_t, a_U, a_L) + \alpha V_U(s_t, a_U), \quad (19)$$

$$U_L(s_t, a_U, a_L) = C(s_t, a_U, a_L) + \beta_p V_L(s_t, a_L). \quad (20)$$

It then applies iterative logit responses over the candidate grid. We call it QRE-style because it is an iterative soft-response approximation rather than an exact analytical QRE solver [9].

Because unrestricted QRE sampled many low-compatibility pairs, we use a compatibility-filtered variant. The QRE joint policy is renormalized over only the top  $N$  candidate pairs by shared compatibility  $C$ . Duration-aware validation selected:

$$\alpha = \beta_p = 0.5, \quad \lambda = 0.5, \quad N = 20. \quad (21)$$

We denote this setting as QRE.

4) *Soft Stackelberg Generation*: Stackelberg generation represents an asymmetric leader–follower interaction. This is musically motivated because one voice may provide a context that the other voice responds to, similar to melody-led or bass-led writing. We test whether such role structure improves coordination beyond simultaneous soft response.

Stackelberg generation uses a leader–follower decision structure [10]. We tested both upper-leading and lower-leading variants. Duration-aware validation selected lower-leading soft Stackelberg:

$$\text{leader} = \text{lower}, \quad \beta_{stk} = 0.75, \quad \alpha = \beta_p = 0.5. \quad (22)$$

We denote this setting as Lower STK. The full validation sweep reveals different strengths across Stackelberg configurations: lower-leading  $\beta_{stk} = 1.0$  obtains the lowest harmonic interval-class JS divergence, while upper-leading  $\beta_{stk} = 0.75$  obtains the lowest voice-distance JS divergence. We nevertheless select lower-leading  $\beta_{stk} = 0.75$  because it provides the best overall balance across vertical similarity, mean voice spacing, duration behavior, pitch entropy, and local pattern diversity.

5) *Pure Nash Ablation*: Pure Nash generation is included as a strict equilibrium ablation. It selects a pair only if neither voice would prefer to change its own token while the other voice remains fixed. This tests whether local strategic stability is beneficial for music generation.

The pure Nash ablation uses the same payoff matrices as QRE and Stackelberg. A candidate pair  $(a_U^i, a_L^j)$  is a pure Nash equilibrium if neither voice can improve its own utility by unilaterally deviating:

$$U_U(i, j) \geq U_U(i', j), \quad \forall i', \quad (23)$$

$$U_L(i, j) \geq U_L(i, j'), \quad \forall j'. \quad (24)$$

When multiple pure equilibria exist, we use welfare tie-breaking. If no equilibrium exists, we fall back to welfare maximization. Duration-aware validation selected  $\alpha = \beta_p = 1.0$  among the tested Nash settings, but Nash remains an ablation rather than a competitive generator [8].

## V. EXPERIMENTAL SETUP

### A. Training

The duration-aware MLP is trained with context length 16, embedding dimension 32, hidden dimension 256, dropout 0.15, and duration feature dimension 6. The duration-aware checkpoint reaches its best validation loss at epoch 11:

$$\text{best validation loss} = 2.4662. \quad (25)$$

Later epochs overfit, so all duration-aware generation uses the best checkpoint.

### B. Generation Length

Each generated sample begins with a real 16-step seed and then generates 128 additional sixteenth-note steps. For evaluation and listening stimuli, the seed prefix is removed:

$$128 \text{ generated steps} \div 16 \text{ steps/bar} = 8 \text{ bars}. \quad (26)$$

At 90 BPM in 4/4, each listening clip is approximately 21.3 seconds. The rendered videos use black screens to avoid visual bias.

### C. Objective Metrics

We compare generated continuations against the held-out real corpus using external music metrics. Lower JS divergences and scalar gaps indicate closer match to the real distribution. Harmonic interval-class JS is computed from the 12-bin distribution of  $|s - b| \bmod 12$  over sounding soprano–bass pairs, whereas voice-distance JS uses the signed soprano–bass spacing  $s - b$  and therefore retains register information. The remaining corpus-similarity metrics include melodic interval JS, pitch-class JS, hold-ratio gap, pitch-entropy gap, unique 3-gram ratio gap, and mean voice-distance gap. Diagnostic and diversity metrics include dissonance rate, excessive leap rate, pitch entropy, and unique 3-gram ratio.

Dissonance rate is computed over timesteps where both voices are sounding after detokenizing HOLD tokens. We compute the vertical interval class modulo 12 and count interval classes outside the consonant set  $\{0, 3, 4, 7, 8, 9\}$  as dissonant. This set corresponds to unison/octave, minor/major thirds, perfect fifths, and minor/major sixths; perfect fourths are therefore treated as dissonant in this diagnostic metric.

Excessive leap rate is computed from consecutive non-rest melodic transitions within each voice after HOLD detokenization. A transition is counted as excessive when the absolute pitch difference is greater than 12 semitones. Thus, leaps larger than an octave are counted, while an exact octave is not.

Internal game metrics such as regret or equilibrium existence are treated only as solver diagnostics. They are not used as direct evidence of musical quality.

#### D. Final-Test Protocol and Paired Uncertainty Analysis

Solver hyperparameters are selected only on the validation split and are frozen before final testing. The final objective evaluation uses all 77 held-out test chorales. For each chorale, every method receives the same 16-step real seed and generates a matched 128-step continuation; the seed prefix is removed before scoring.

To quantify cross-chorale uncertainty, we use 10,000 paired percentile-bootstrap resamples over the 77 matched test chorales. Each bootstrap replicate samples 77 chorale indices with replacement and applies the same resampled indices to every method, after which the original corpus-level statistics and comparison metrics are recomputed against the fixed real test corpus. We report 95% confidence intervals for

$$\Delta = \text{metric}_{\text{method}} - \text{metric}_{\text{No-game}}. \quad (27)$$

Because every reported objective metric is lower-is-better,  $\Delta > 0$  favors No-game and  $\Delta < 0$  favors the compared method. A confidence interval containing zero is treated as an uncertain method difference rather than evidence that either method is superior.

#### VI. OBJECTIVE VALIDATION RESULTS

Table I summarizes key duration-aware validation results. The real validation reference is included where the raw metric is meaningful.

For compactness, the tables use the following method labels: No-game joint denotes the compatibility-only baseline; QRE denotes the selected compatibility-filtered QRE setting; Lower STK denotes the selected lower-leading soft Stackelberg setting; Upper STK denotes the upper-leading Stackelberg leader ablation, not the selected main Stackelberg method.

The validation results show three important patterns. First, No-game joint is a strong baseline: it closely matches the real hold ratio and has high pitch entropy and diversity. Second, QRE matches duration behavior well, but its harmonic interval-class and voice-distance distributions remain weaker than the strongest alternatives and its dissonance rate is high. Third, Stackelberg can improve vertical distribution metrics, but the best configuration depends on which aspect of vertical structure is emphasized. Nash performs worst on diversity and hold behavior, showing that strict equilibrium selection is not musically desirable in this local generation setting.

Table II highlights why Lower STK with  $\beta_{stk} = 0.75$  was selected as the balanced Stackelberg setting rather than selecting a configuration by a single vertical metric.

No single Stackelberg configuration dominates all validation metrics. Lower-leading  $\beta_{stk} = 1.0$  gives the lowest harmonic interval-class JS divergence, while upper-leading  $\beta_{stk} = 0.75$  gives the lowest voice-distance JS divergence. The selected lower-leading  $\beta_{stk} = 0.75$  setting is slightly weaker on those individual metrics but has by far the smallest mean voice-distance gap and the smallest unique 3-gram and pitch-entropy gaps among the three configurations in Table II. We therefore use it as the balanced Stackelberg setting and retain upper-leading  $\beta_{stk} = 0.75$  as a leader ablation.

#### VII. FINAL OBJECTIVE TEST RESULTS

After selecting solver settings on the validation split, we perform the final objective evaluation on all 77 held-out test chorales. Each method therefore contributes 77 matched 128-step continuations, with the same 16-step real prefix removed before scoring. The evaluated methods are No-game joint, QRE, the validation-selected Lower STK, upper-leading Stackelberg with  $\beta_{stk} = 0.75$  as a leader ablation, and Nash as a pure-equilibrium ablation.

The full-test point estimates redistribute several apparent “winners” from the earlier 30-continuation analysis. No-game joint retains the lowest harmonic interval-class and voice-distance JS values, which are the two metrics most directly tied to vertical soprano–bass coordination. QRE now has the lowest point estimates for both melodic-interval divergences and hold-ratio gap. Lower STK has the lowest pitch-class JS and pitch-entropy gap, whereas Upper STK has the lowest mean voice-distance and unique 3-gram gaps. These shifts already indicate that single-metric rankings can be sensitive to which test chorales are sampled.

Tables V and VI therefore report paired-bootstrap differences relative to No-game. Bold entries have 95% confidence intervals that exclude zero. Positive values favor No-game because lower metric values are better; negative values favor the compared method.

The bootstrap analysis changes the interpretation from a leaderboard to a set of method-specific effects. QRE is reliably farther from the real corpus than No-game on harmonic interval-class, voice-distance, pitch-class, pitch-entropy, and local 3-gram matching. Although QRE has the lowest point estimates for hold-ratio and both melodic-interval divergences, the paired intervals for those three apparent advantages all include zero. Thus, QRE changes temporal and melodic behavior, but the full-test analysis does not establish a reliable improvement over No-game on those metrics.

Stackelberg shows a different pattern. Both Lower and Upper STK have reliably lower unique 3-gram gaps than No-game under paired resampling, giving the clearest robust positive game-theoretic effect in the objective study. By contrast, the spacing story from the earlier 30-continuation analysis does not generalize: the lowest mean voice-distance gap shifts from Lower STK to Upper STK in the 77-chorale point estimates, but both paired intervals relative to No-game are wide and include zero. Lower STK’s pitch-class and pitch-entropy point-estimate advantages are likewise uncertain. Upper STK is especially informative because its harmonic interval-class and voice-distance differences from No-game are both very small and their paired intervals include zero, while its local-diversity advantage is reliable. This is evidence of a coordination–diversity trade-off rather than uniform superiority.

Nash remains the clearest failure mode. Its hold ratio is 0.966873 compared with 0.756217 for real music, and its unique 3-gram ratio is only 0.082973 compared with 0.569775 for real music. Eight of the nine paired metric differences

TABLE I  
DURATION-AWARE VALIDATION RESULTS FOR SELECTED AND DIAGNOSTIC METHODS

| Method           | Harmonic IC JS ↓ | Voice Dist. JS ↓ | Hold Ratio | Dissonance | Pitch Entropy | Unique 3-gram |
|------------------|------------------|------------------|------------|------------|---------------|---------------|
| Real             | –                | –                | 0.767465   | 0.149649   | 3.350719      | 0.553407      |
| No-game joint    | 0.014123         | 0.019738         | 0.772266   | 0.201477   | 3.371686      | 0.692593      |
| QRE              | 0.028802         | 0.033620         | 0.768620   | 0.313163   | 3.241633      | 0.735450      |
| Lower STK (sel.) | 0.010160         | 0.015468         | 0.808724   | 0.193582   | 3.304204      | 0.600529      |
| Upper STK (abl.) | <b>0.008873</b>  | <b>0.014099</b>  | 0.798568   | 0.199474   | 3.410057      | 0.629630      |
| Nash             | 0.078304         | 0.118107         | 0.972266   | 0.093571   | 2.924298      | 0.068783      |

TABLE II  
DURATION-AWARE STACKELBERG VALIDATION TRADE-OFFS

| Metric                       | Lower STK $\beta = .75$ (sel.) | Lower STK $\beta = 1.0$ | Upper STK $\beta = .75$ |
|------------------------------|--------------------------------|-------------------------|-------------------------|
| Harmonic interval-class JS ↓ |                                | 0.010160                | <b>0.008134</b>         |
| Voice-distance JS ↓          |                                | 0.015468                | <b>0.014099</b>         |
| Hold-ratio gap ↓             |                                | 0.041259                | <b>0.031102</b>         |
| Mean voice-distance gap ↓    |                                | <b>0.027515</b>         | 0.114056                |
| Unique 3-gram gap ↓          |                                | <b>0.047122</b>         | 0.098381                |
| Pitch-entropy gap ↓          |                                | <b>0.046515</b>         | 0.049665                |

TABLE III  
FULL 77-CHORALE FINAL OBJECTIVE TEST: CORPUS-SIMILARITY METRICS AND SCALAR GAPS. LOWER IS BETTER.

| Metric                      | Nash     | No-game joint   | QRE             | Lower STK (sel.) | Upper STK       |
|-----------------------------|----------|-----------------|-----------------|------------------|-----------------|
| Bass melodic interval JS    | 0.067153 | 0.005900        | <b>0.004951</b> | 0.006242         | 0.005623        |
| Harmonic interval-class JS  | 0.084715 | <b>0.007332</b> | 0.028789        | 0.012357         | 0.007486        |
| Hold-ratio gap              | 0.210656 | 0.014786        | <b>0.010321</b> | 0.034367         | 0.031831        |
| Mean voice-distance gap     | 0.318951 | 0.187996        | 0.256309        | 0.372054         | <b>0.064271</b> |
| Pitch-class JS              | 0.047688 | 0.001124        | 0.005032        | <b>0.000886</b>  | 0.002841        |
| Pitch-entropy gap           | 0.337981 | 0.027990        | 0.129482        | <b>0.004326</b>  | 0.017666        |
| Soprano melodic interval JS | 0.043195 | 0.006092        | <b>0.004903</b> | 0.005357         | 0.005524        |
| Unique 3-gram ratio gap     | 0.486802 | 0.116991        | 0.167599        | 0.081328         | <b>0.077720</b> |
| Voice-distance JS           | 0.100913 | <b>0.009404</b> | 0.033197        | 0.014316         | 0.009862        |

TABLE IV  
FULL 77-CHORALE FINAL OBJECTIVE TEST: DIAGNOSTIC AND DIVERSITY METRICS

| Metric                | Real Test | Nash      | No-game joint | QRE       | Lower STK (sel.) | Upper STK |
|-----------------------|-----------|-----------|---------------|-----------|------------------|-----------|
| Hold ratio            | 0.756217  | 0.966873  | 0.771002      | 0.766538  | 0.790584         | 0.788048  |
| Voice crossing rate   | 0.000000  | 0.000000  | 0.000000      | 0.000000  | 0.000000         | 0.000000  |
| Dissonance rate       | 0.141833  | 0.020083  | 0.194922      | 0.309598  | 0.227478         | 0.196762  |
| Parallel perfect rate | 0.009980  | 0.003606  | 0.008875      | 0.010507  | 0.009309         | 0.011985  |
| Excessive leap rate   | 0.000213  | 0.000000  | 0.000925      | 0.000567  | 0.000978         | 0.001029  |
| Pitch entropy         | 3.394889  | 3.056909  | 3.422879      | 3.265407  | 3.390563         | 3.377224  |
| Unique 3-gram ratio   | 0.569775  | 0.082973  | 0.686766      | 0.737374  | 0.651103         | 0.647495  |
| Mean voice distance   | 19.650117 | 19.969067 | 19.462121     | 19.393808 | 19.278063        | 19.714388 |

TABLE V  
PAIRED BOOTSTRAP DIFFERENCES RELATIVE TO NO-GAME: CORPUS-SIMILARITY METRICS. ENTRIES ARE  $\Delta$  [95% CI], WHERE  $\Delta$  = METHOD – NO-GAME. BOLD INTERVALS EXCLUDE ZERO.

| Method    | Bass MI JS                    | Harmonic IC JS                | Pitch-class JS                | Soprano MI JS                 | Voice-distance JS             |
|-----------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| QRE       | –0.00095 [–0.00245, 0.00060]  | +0.02146 [+0.01646, +0.02633] | +0.00391 [+0.00061, +0.00867] | –0.00119 [–0.00250, 0.00019]  | +0.02379 [+0.01793, +0.02896] |
| Lower STK | +0.00034 [–0.00127, +0.00203] | +0.00502 [+0.00010, +0.01039] | –0.00024 [–0.00217, +0.00192] | –0.00074 [–0.00250, +0.00111] | +0.00491 [–0.00061, +0.01149] |
| Upper STK | –0.00028 [–0.00194, +0.00149] | +0.00015 [–0.00345, +0.00399] | +0.00172 [–0.00060, +0.00517] | –0.00057 [–0.00217, +0.00101] | +0.00046 [–0.00395, +0.00538] |
| Nash      | +0.06125 [+0.05312, +0.06975] | +0.07738 [+0.05918, +0.11936] | +0.04656 [+0.03249, +0.08412] | +0.03710 [+0.03168, +0.04295] | +0.09151 [+0.07596, +0.13890] |

TABLE VI  
PAIRED BOOTSTRAP DIFFERENCES RELATIVE TO NO-GAME: DURATION AND DIVERSITY GAPS. ENTRIES ARE  $\Delta$  [95% CI]. BOLD INTERVALS EXCLUDE ZERO.

| Method    | Hold-ratio gap                | Mean voice-distance gap       | Pitch-entropy gap             | Unique 3-gram gap             |
|-----------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| QRE       | –0.00446 [–0.01126, +0.00213] | +0.06831 [–0.33361, +0.42624] | +0.10149 [+0.07100, +0.16180] | +0.05061 [+0.03195, +0.06988] |
| Lower STK | +0.01958 [+0.01096, +0.02800] | +0.18406 [–0.28255, +0.58917] | –0.02366 [–0.03864, +0.03539] | –0.03566 [–0.05628, –0.01525] |
| Upper STK | +0.01705 [+0.00893, +0.02521] | –0.12372 [–0.42956, +0.28056] | –0.01032 [–0.03559, +0.04820] | –0.03927 [–0.06164, –0.01680] |
| Nash      | +0.19587 [+0.18425, +0.20602] | +0.13095 [–0.33863, +1.13360] | +0.30999 [+0.28675, +0.50023] | +0.36981 [+0.34651, +0.39403] |

relative to No-game exclude zero in the direction favoring No-game; only mean voice-distance gap remains too variable to distinguish reliably. The collapse therefore persists across the complete held-out test split rather than being an artifact of the original 30-continuation sample.

Overall, the final objective experiment supports a more specific conclusion than the preliminary test. Game-theoretic inference does measurably change the generated distribution, but its benefits are selective. Stackelberg provides a robust improvement in local diversity matching, QRE does not show a reliable advantage on its apparent temporal/melodic strengths, and strict Nash is consistently degenerate. No-game remains a strong vertical-coordination baseline, but it is no longer accurate to describe it as dominating every distributional aspect of the generation.

## VIII. LISTENING TEST

### A. Participants and Recruitment

Participants were recruited online through an open voluntary call. Because recruitment was open rather than drawn from a defined sampling frame, the study should be interpreted as an online convenience sample rather than a probability-based random sample. No eligibility screening based on musical training, classical-music familiarity, or professional music experience was applied. For consistency with the perceptual analyses reported throughout this paper, we use the first 47 completed responses; responses submitted later to the online form are not included in the results reported here.

Musical background was self-reported before the listening trials. Of the 47 participants, 23 (48.9%) reported no formal music training, 7 (14.9%) reported less than one year, 9 (19.1%) reported one to three years, and 8 (17.0%) reported more than four years. Thus, 24 participants (51.1%) had at least some formal training, while nearly half had none. Participants also rated their familiarity with classical music on a 1–5 scale. The distribution was 11, 14, 11, 7, and 4 participants at levels 1 through 5, respectively; the mean rating was 2.55 and the median was 2. Overall, the sample therefore contains a mixture of musically untrained and trained listeners, with self-reported classical-music familiarity concentrated toward the low-to-moderate range. These background variables are reported descriptively and are not used to filter, weight, or stratify the hypothesis tests.

### B. Design and Procedure

To evaluate perceptual quality, we conducted a blind A/B listening test using duration-aware generated samples. The listening-test methods were chosen from the validation-stage selections before the expanded 77-chorale final-test analysis: QRE is the selected compatibility-filtered soft-response setting, and Lower STK is the validation-selected balanced Stackelberg representative. The listening study therefore does not select a Stackelberg variant after inspecting the final-test outcomes.

The three method comparisons were:

- No-game joint vs. Lower STK,

- No-game joint vs. QRE,
- Lower STK vs. QRE.

Each comparison used four matched seeds, giving 12 A/B trials in total. The two clips within a trial were generated from the same real seed, so the comparison changes the generation rule while holding the musical starting context fixed. As in the objective evaluation, the seed prefix itself was removed from the listening stimulus. Each continuation contains 128 generated sixteenth-note steps, corresponding to 8 bars; at 90 BPM in 4/4, each clip is approximately 21.3 seconds long. Both clips were rendered using the same piano timbre and tempo and presented as black-screen videos to minimize visual cues. The method identities associated with Clip A and Clip B were hidden from participants.

For each trial, participants answered three perceptual questions:

- 1) Which clip has better coordination between the upper and lower voices?
- 2) Which clip sounds more natural and coherent overall?
- 3) Overall, which clip do you prefer?

The response options were A, B, and no clear preference. The questionnaire also provided an optional free-text field in which participants could explain their choice or describe audible differences. These comments were collected as qualitative feedback but were not used in the quantitative hypothesis tests reported below.

### C. Statistical Analysis

Each method comparison contains four matched trials. Since each participant answers four trials within the same comparison, pooled trial-level votes are not independent. We therefore report pooled votes descriptively, but perform statistical testing at the participant level.

For each participant, comparison, and question, the four responses are encoded as:

$$r = \begin{cases} +1, & \text{if Method 1 is preferred,} \\ -1, & \text{if Method 2 is preferred,} \\ 0, & \text{if there is no clear preference.} \end{cases} \quad (28)$$

The four values are summed into a participant-level preference score. A participant with score  $> 0$  is classified as preferring Method 1, score  $< 0$  as preferring Method 2, and score  $= 0$  as a participant-level tie. We then apply a two-sided exact binomial test to the numbers of participants preferring each method, excluding participant-level ties. We report the uncorrected  $p$ -values and use  $p < .05$  only as a nominal exploratory threshold. Because nine participant-level tests are reported, these results should not be interpreted as confirmatory; under a Holm correction across the nine tests, only the No-game joint vs. Lower STK coherence comparison remains below 0.05.

### D. Listening Test Results

Table VIII shows that No-game joint was favored over Lower STK in all three questions. At the nominal uncorrected  $p < .05$  threshold, the participant-level tests favor No-game

TABLE VII  
MUSICAL BACKGROUND OF THE 47 LISTENING-TEST PARTICIPANTS

| Measure                                          | Response category | <i>n</i> | Percentage |
|--------------------------------------------------|-------------------|----------|------------|
| Formal music training                            | None              | 23       | 48.9%      |
|                                                  | Less than 1 year  | 7        | 14.9%      |
|                                                  | 1–3 years         | 9        | 19.1%      |
|                                                  | More than 4 years | 8        | 17.0%      |
| Classical-music familiarity<br>(1–5 self-rating) | 1                 | 11       | 23.4%      |
|                                                  | 2                 | 14       | 29.8%      |
|                                                  | 3                 | 11       | 23.4%      |
|                                                  | 4                 | 7        | 14.9%      |
|                                                  | 5                 | 4        | 8.5%       |

joint for coordination and coherence/naturalness; the coherence result is also the only listening-test comparison that remains below 0.05 after Holm correction across all nine tests. Overall preference shows the same direction but does not reach the nominal threshold.

Table IX shows that No-game joint was favored over QRE on all three questions. Overall preference reaches the nominal uncorrected  $p < .05$  threshold, while coordination and coherence show the same direction without reaching that threshold. This comparison does not remain significant after correction across the nine reported tests.

Table X shows no statistically significant difference between Stackelberg and QRE, although the pooled and participant-level trends generally favored Stackelberg.

## IX. DISCUSSION

### A. Learned Compatibility Remains a Strong Vertical Baseline

The expanded test set changes the interpretation of No-game joint in an important way. It is no longer the point-estimate winner on bass melodic-interval or pitch-class JS, but it remains best on the two metrics most directly tied to vertical soprano–bass coordination: harmonic interval-class and voice-distance JS. Against QRE, both advantages are robust under paired bootstrap. Against Upper STK, however, the corresponding confidence intervals include zero. Learned pair compatibility is therefore clearly strong, but the full-test results do not support the stronger claim that it uniformly dominates every strategic alternative on every aspect of vertical behavior.

This distinction matters for the central research question. An improvement over independent generation would not by itself demonstrate the value of game solving, because learned pair compatibility already provides joint coordination. The full-test results instead identify where strategic structure adds a measurable effect beyond that strong baseline.

### B. Game Solvers Produce Selective Rather Than Uniform Gains

QRE produces the clearest behavioral shift away from No-game. Its point estimates are lowest for hold-ratio gap and both melodic-interval divergences, but all three paired confidence intervals relative to No-game include zero. At the same time,

QRE is reliably worse on both main vertical JS metrics and on several distributional or diversity gaps. The correct interpretation is therefore not that QRE reliably improves duration or melody, but that its soft-response rule redistributes behavior toward those dimensions while sacrificing other aspects of corpus matching.

Stackelberg yields the strongest positive game-theoretic result. Both leader variants reliably reduce the unique 3-gram ratio gap relative to No-game, showing a repeatable improvement in local-diversity matching. However, the earlier spacing claim is not robust: Lower STK’s strong mean voice-distance gap in the preliminary 30-continuation analysis does not generalize to the full test set, and neither Stackelberg variant has a reliable paired advantage over No-game on mean spacing. Upper STK is particularly revealing because it preserves vertical-distribution statistics close to No-game while obtaining a reliable local-diversity improvement. Thus, Stackelberg changes the coordination–diversity balance rather than simply improving all musical statistics.

Pure Nash still provides the clearest cautionary result. Its extreme hold ratio and very low unique 3-gram ratio persist over all 77 test chorales, and almost every paired metric difference strongly favors No-game. Local strategic stability can therefore be achieved through musically conservative sustain rather than through useful long-term coordination.

### C. Objective Metrics and Human Preference Provide Complementary Evidence

The listening test used the validation-selected Lower STK configuration rather than a variant chosen after inspecting final-test results. This is important because the expanded objective test revises some of the preliminary ranking-specific conclusions: Lower STK is no longer the mean-spacing winner on the full test set. Nevertheless, it remains a meaningful perceptual comparison because it was the pre-specified balanced Stackelberg representative and because its local 3-gram diversity advantage over No-game is reliable under paired bootstrap.

The perceptual result is therefore not invalidated by the revised objective ranking; instead, it becomes more informative. Lower STK moves local-diversity statistics closer to the real corpus, yet participants still favor No-game joint for perceived

TABLE VIII  
LISTENING TEST: NO-GAME JOINT VS. LOWER STK (SELECTED)

| Question     | M1 Votes | M2 Votes | No Pref. | M1 Part. | M2 Part. | $p$    |
|--------------|----------|----------|----------|----------|----------|--------|
| Coordination | 89       | 64       | 35       | 26       | 12       | 0.0336 |
| Coherence    | 94       | 66       | 28       | 24       | 7        | 0.0033 |
| Overall      | 80       | 59       | 49       | 20       | 9        | 0.0614 |

TABLE IX  
LISTENING TEST: NO-GAME JOINT VS. QRE

| Question     | M1 Votes | M2 Votes | No Pref. | M1 Part. | M2 Part. | $p$    |
|--------------|----------|----------|----------|----------|----------|--------|
| Coordination | 81       | 61       | 46       | 22       | 12       | 0.1214 |
| Coherence    | 86       | 64       | 38       | 19       | 8        | 0.0522 |
| Overall      | 77       | 51       | 60       | 19       | 7        | 0.0290 |

TABLE X  
LISTENING TEST: LOWER STK (SELECTED) VS. QRE

| Question     | M1 Votes | M2 Votes | No Pref. | M1 Part. | M2 Part. | $p$    |
|--------------|----------|----------|----------|----------|----------|--------|
| Coordination | 86       | 70       | 32       | 19       | 14       | 0.4869 |
| Coherence    | 96       | 67       | 25       | 24       | 13       | 0.0989 |
| Overall      | 79       | 62       | 47       | 20       | 13       | 0.2962 |

coherence, with the coherence comparison remaining below 0.05 after Holm correction. This shows that improved local distributional matching does not necessarily imply improved perceived coherence. Objective statistics and human judgments are measuring overlapping but non-identical aspects of musical quality.

The participant background also helps contextualize this result. The listening panel was not restricted to trained musicians: 48.9% reported no formal music training, while the remainder covered a range from less than one year to more than four years of training. Classical-music familiarity was similarly mixed and centered below the midpoint of the 1–5 scale. The reported preference therefore reflects a mixed-experience online listener group rather than an expert-panel judgment. We do not claim that musical expertise has no effect on preference; the present sample is not designed or powered for subgroup inference, so experience-related differences remain a question for future work.

#### D. Why This Is Still a Useful Result

The expanded experiment strengthens the scientific contribution because it moves the paper away from a simple method leaderboard. The decision rule clearly matters: QRE, Stackelberg, and Nash produce distinct and repeatable distributional signatures. What does not hold is the stronger claim that more strategic structure is uniformly better. The most defensible conclusion is that game-theoretic inference creates selective trade-offs. In particular, Stackelberg can improve local diversity without a reliable vertical-distribution advantage, while strict Nash can optimize local stability in a musically degenerate way. These findings motivate future systems that combine learned compatibility with softened strategic interaction rather than optimizing equilibrium structure for its own sake.

## X. LIMITATIONS

This project has several limitations. First, the utility model is a relatively small MLP trained on a simplified state representation. Even with duration-aware features, it does not explicitly model key, phrase structure, long-range form, or tonal function. Second, the game is solved myopically at each timestep; there is no lookahead or planning over future musical consequences. Third, the objective bootstrap quantifies uncertainty across the 77 held-out chorales but does not capture model-training uncertainty or repeated stochastic generations from multiple random seeds within the same chorale, because the final test uses one matched continuation per chorale per method. Fourth, the listening test uses an online convenience sample of 47 volunteers rather than a probability-based or expertise-stratified sample. Musical experience is mixed and was not used as an inclusion criterion or balancing variable, so the perceptual results should be interpreted as judgments from a heterogeneous volunteer sample rather than as estimates for a representative population or an expert-only panel. The study also includes only three method pairings; Holm correction is applied across the nine participant-level tests, and only the No-game vs. Lower STK coherence comparison remains below 0.05. Finally, the stimuli contain only 8-bar generated continuations, so long-form musical coherence is not evaluated. The findings should therefore be interpreted as evidence about inference-time decision rules under this fixed model, dataset, and continuation length rather than as a universal ranking of game-theoretic music generators.

## XI. CONCLUSION

This study provides a controlled evaluation of game-theoretic decision rules for two-voice symbolic music generation. Using a shared duration-aware model and candidate

grid, we evaluate all 77 held-out test chorales and use paired bootstrap resampling to separate robust effects from unstable point-estimate rankings.

The expanded results revise several preliminary conclusions without changing the central thesis. No-game joint remains a strong vertical-coordination baseline, with the lowest harmonic interval-class and voice-distance JS point estimates. QRE shifts hold and melodic behavior, but its apparent advantages over No-game on those metrics are not reliable under paired resampling and it remains clearly weaker on vertical-distribution matching. Soft Stackelberg provides the strongest positive strategic effect: both leader variants reliably improve local 3-gram diversity matching, although the earlier mean-spacing advantage of Lower STK does not generalize to the full test set. Pure Nash remains robustly collapsed toward sustained, low-diversity output.

The listening test complements these objective findings. The validation-selected Lower STK configuration improves local-diversity matching, yet listeners still favor No-game joint for perceived coherence. This discrepancy reinforces the main conclusion: game-theoretic inference meaningfully changes musical behavior, but its benefits are selective, metric-dependent, and not automatically perceptual. Future systems should therefore use strategic structure as a controllable inductive bias, combined with learned compatibility and temporal awareness, rather than treating stronger equilibrium structure as an objective in itself.

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